



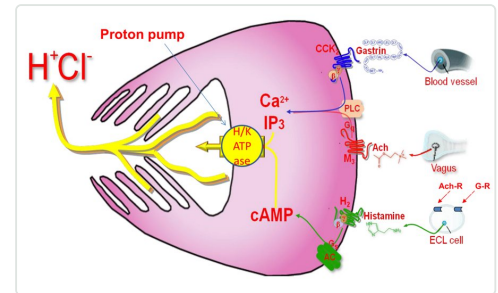
BEFORE YOU READ — ANSWER FROM MEMORY

1. Which three signals stimulate the parietal-cell proton pump, and which two messengers do they use?
2. Why is a PPI taken ~1 hour before a meal, and why does one dose act for up to 24 h despite a ~1.5 h half-life?
3. Which drug must not be given together with antacids or H₂ blockers, and why?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 How the parietal cell makes acid

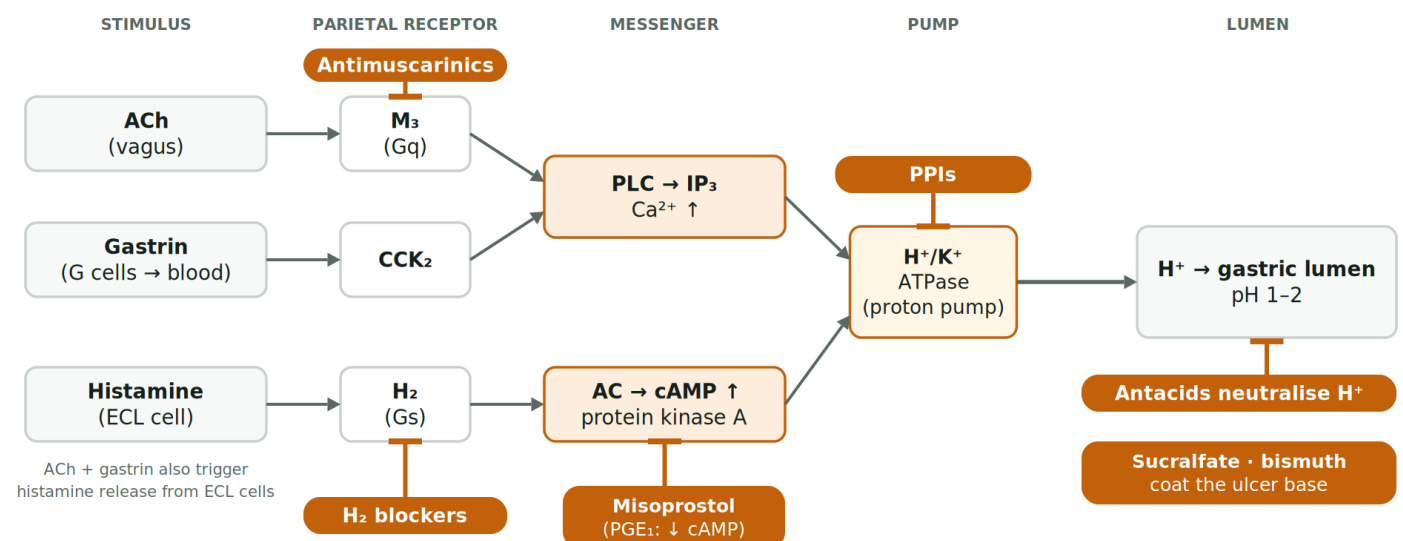
Parietal cell	Body & fundus. Secretes HCl and intrinsic factor (needed for vitamin B12 absorption). Has receptors for gastrin , ACh (M₃) , histamine (H₂)
G cell	Endocrine cell of the gastric antrum and duodenum → secretes gastrin
ECL cell	In the fundus ; makes, stores and releases histamine in response to ACh and gastrin
Convergence	ACh and gastrin act on ECL cells as well as parietal cells ; the histamine released then stimulates the pump via H₂
Basal acid output	Minimum HCl produced per unit time; normal adult 2-5 mEq/hr
Defence	Mucus-secreting cells make mucus + HCO₃⁻ (trapped in the mucus layer). PGE₂ and PGI₂ ↓ acid (↓ adenylate cyclase) but ↑ mucus and HCO ₃ ⁻ and dilate mucosal vessels



Lecture slide — mechanism of acid production

Signal	Receptor → messenger	Result
ACh (vagus)	M₃ → Gq → PLC → IP ₃ / Ca ²⁺ (Ca ²⁺ -dependent protein kinase)	Activates the H⁺/K⁺ ATPase
Gastrin (blood)	CCK₂ → PLC → IP ₃ / Ca ²⁺	
Histamine (ECL)	H₂ → Gs → adenylate cyclase → cAMP → protein kinase A	Tubulovesicles fuse with the canalicular membrane → H⁺ released into lumen

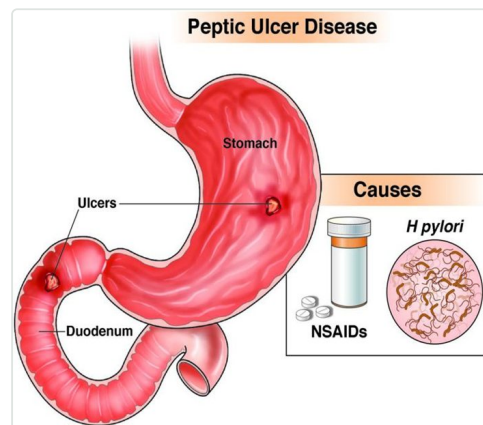
Visual anchor — where each drug class acts



Bars (—) mark the step each drug class blocks or opposes; antacids and coating agents act in the lumen / on the ulcer. Only mechanisms given in the lecture are shown.

2 Peptic ulcer disease and H. pylori

Acid-related diseases	Dyspepsia, GERD , PUD , upper GI bleeding — mostly from ↑ acid secretion
PUD	Chronic, remitting and relapsing (esp. duodenal), lasting years; ~10% of people in western countries; DU : GU = 4-5 : 1
Causes	NSAID use · H. pylori · ↑ HCl secretion · inadequate mucosal defence
Treatment goals	Pain relief · ulcer healing · prevent complications (bleeding, perforation) · prevent relapse



Lecture slide — ulcer sites and causes

H. pylori regimen	Drugs (2-week course)
Triple therapy	PPI + (metronidazole or amoxicillin) + clarithromycin
Quadruple therapy	Bismuth subsalicylate + metronidazole + tetracycline + PPI

Any PUD patient (duodenal or gastric ulcer) who is H. pylori-infected needs antimicrobial treatment. Eradication is ~90% with these regimens (the lecture quotes 80-90% and ≥90% on consecutive slides).

3 Acid suppressants — PPIs vs H₂ blockers

	Proton pump inhibitors (PPIs)	H ₂ receptor antagonists
Drugs	omeprazole, lansoprazole, rabeprazole, pantoprazole, esomeprazole	cimetidine, ranitidine, famotidine, nizatidine
Target & binding	H⁺/K⁺ ATPase (final step of acid secretion). Prodrugs : activated in the canaliculus, form a covalent bond with a cysteine residue → irreversible	H₂ receptor : selective, competitive, fully reversible ; ↓ cAMP → ↓ acid
Potency	Superior to H ₂ blockers for acid suppression and ulcer healing	>90% inhibition of basal, food-stimulated and nocturnal acid; use has fallen since PPIs
Kinetics	Oral. Food ↓ bioavailability by 50% → give ~1 h before a meal . t _{1/2} ≈ 1.5 h but effect lasts up to 24 h (~18 h to resynthesise the pump)	First-pass → ~50% bioavailability (cimetidine, ranitidine, famotidine); nizatidine ~100% , cleared mainly by kidney. ↓ dose in moderate-severe renal (± severe hepatic) insufficiency and in the elderly
Uses	Peptic ulcer healing · long-term Zollinger-Ellison (gastrin-secreting tumour, HCl hypersecretion) · GERD	Peptic ulcers · acute stress ulcer (given IV) · GERD (PPIs now preferred)
Adverse effects	Generally well tolerated. Long-term use → subnormal vitamin B12 (acid is needed to release B12 from food)	Extremely safe. Diarrhoea, headache, fatigue, muscle pain, constipation; CNS: confusion, hallucinations. Cimetidine : male sexual dysfunction, gynaecomastia . Ranitidine: minimal
Interactions	None stated in the lecture	Cimetidine inhibits CYP450 → ↑ effect of warfarin, diazepam, phenytoin, quinidine, carbamazepine, theophylline, imipramine

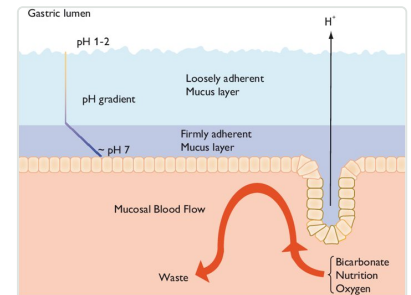
4 Prostaglandin analogue, antimuscarinics and mucosal protectives

Drug	Mechanism	Use	Adverse effects / watch-out
Misoprostol synthetic PGE ₁	Binds a PG receptor on parietal cells → ↓ histamine-stimulated cAMP → modest acid inhibition ; ↑ mucus and HCO ₃ ⁻ (cytoprotective). Short action (~3 h) → less 24-h inhibition than H ₂ blockers	With PPIs, prevents NSAID-induced gastric ulcers . Weaker than H ₂ blockers/PPIs for acute ulcer treatment	Uterine contractions → contraindicated in pregnancy . Dose-related diarrhoea, nausea limit use

Pirenzepine / dicyclomine	Antimuscarinics: pirenzepine = selective M₁ blocker; dicyclomine blocks M₁, M₂, M₃	PUD and Zollinger-Ellison, esp. patients refractory to standard therapy	Arrhythmias, dry mouth, constipation, urinary retention → limit use
Sucralfate	Sucrose + sulfated Al(OH) ₃ . In acid forms a viscous paste binding ulcers/erosions up to 6 h (physical barrier); ↑ mucosal PG and HCO ₃ ⁻	Heals duodenal ulcers ; does not prevent NSAID ulcers and does not heal gastric ulcers	Needs acidic pH → do not give with H ₂ blockers or antacids
Bismuth subsalicylate	Antimicrobial; coats the ulcer base , inhibits pepsin, ↑ mucus	Heals peptic ulcers; component of quadruple therapy	Nausea, vomiting; black tongue and faeces

Why the mucosal barrier works

Barrier	Mucus and epithelial tight junctions restrict back-diffusion of acid and pepsin
pH gradient	Epithelial HCO ₃ ⁻ makes a gradient in the mucus: pH ~7 at the surface → pH 1-2 in the lumen
Blood flow	Carries HCO ₃ ⁻ and nutrients to surface cells; mucosal prostaglandins drive mucus, HCO ₃ ⁻ and blood flow



Lecture slide — mucosal defence

5 Antacids

Action	Weak bases: acid + antacid → water + salt . Partly neutralise gastric acid and inhibit pepsin (inactive at pH > 4)
Onset / duration	Very quick (minutes) but short — about 30-60 min on an empty stomach

Antacid	Key points	Discriminator / use
NaHCO₃ systemic	Potent (pH may rise above 7); soluble and absorbed at high doses → systemic alkalosis ; Na⁺ load worsens oedema and CHF; CO ₂ in stomach → distension, belching, risk of ulcer perforation	Casual heartburn relief; also alkalinises urine and treats acidosis
Mg(OH)₂ non-systemic	Diarrhoea / loose stools when used alone; hyporeflexia, hypotension, cardiac arrest	Non-systemic antacids are insoluble and poorly absorbed → form the chloride salt, which reacts with intestinal HCO ₃ ⁻ → no acid-base disturbance . Mg + Al together cancel each other's bowel effects
Al(OH)₃ non-systemic	Al ³⁺ relaxes smooth muscle → delayed gastric emptying, constipation . Binds phosphate → hypophosphatemia → osteomalacia. In renal failure Al ³⁺ accumulates → encephalopathy, osteoporosis	
CaCO₃	Potent, rapid; liberates CO ₂ (slower than NaHCO ₃) → distension, discomfort	Contraindications (listed after the CaCO ₃ slide): hypertension, heart failure, renal disease

Phosphate binding by Al³⁺ is also used therapeutically: in hyperphosphatemia and phosphate stones.

Antacid interaction	Examples
↑ gastric pH changes bioavailability	Iron, tetracyclines, digoxin, prednisolone, propranolol, quinidine, antimuscarinics
Chelation (insoluble complexes)	Quinolones with Ca ²⁺ , Mg ²⁺ or Al ³⁺ · tetracyclines with Ca ²⁺
Urine alkalisation	↑ excretion of phenobarbital, methotrexate, salicylate

P Must-remember pearls

- **PPIs block the H^+/K^+ ATPase irreversibly** (final step of acid secretion): take ~1 h before a meal; effect lasts up to 24 h; long-term use → ↓ vitamin B12.
- **Cimetidine is the H_2 blocker that inhibits CYP450** (warfarin, phenytoin, diazepam...) and causes gynaecomastia / male sexual dysfunction; H_2 blockers need a dose cut in renal impairment and the elderly.
- **Misoprostol (PGE_1 analogue) + PPI prevents NSAID ulcers** — but it is contraindicated in pregnancy and causes diarrhoea.
- **Sucralfate needs acid** (no H_2 blockers or antacids with it); **bismuth** blackens tongue and faeces.
- **Antacid signatures:** Mg → diarrhoea · Al → constipation + hypophosphatemia · Na → alkalosis and Na^+ load · Ca/Na → CO_2 distension. Their ions **chelate** quinolones and tetracyclines.

Q Test yourself

1. After a large meal, gastrin from the antral G cells and vagal acetylcholine reach the gastric fundus. Which cell converts both signals into a histamine message for the parietal cell?

- A. Antral G cell B. Mucus-secreting cell C. ECL cell D. Parietal cell itself E. Chief cell

2. An ICU patient at risk of acute stress ulcer cannot take tablets. According to the lecture, which class is usually injected intravenously for this?

- A. H_2 antagonists B. Misoprostol C. Sucralfate D. Sodium bicarbonate E. Bismuth subsalicylate

3. A patient with a peptic ulcer takes an antacid. Apart from neutralising acid, which effect further reduces damage to the ulcer?

- A. Irreversible blockade of the proton pump B. Blockade of parietal H_2 receptors
C. Formation of a protective paste that binds the ulcer for up to 6 hours D. Direct stimulation of mucosal prostaglandin synthesis
E. Inhibition of pepsin, which is inactive at a pH above 4

4. A patient's gastric mucosa is deficient in PGE_2 and PGI_2 . Which protective actions is it missing?

- A. ↑ acid secretion (↑ adenylate cyclase), ↑ mucus and HCO_3^- secretion, ↓ mucosal blood flow B. ↓ acid secretion (↓ adenylate cyclase), ↑ mucus and HCO_3^- secretion, ↑ mucosal blood flow
C. ↓ acid secretion (↓ adenylate cyclase), ↓ mucus and HCO_3^- secretion, ↑ mucosal blood flow D. ↑ gastrin release, ↑ mucus secretion, ↓ HCO_3^- secretion, no change in blood flow
E. ↑ pepsin activity and ↑ HCO_3^- secretion, with no effect on acid secretion

5. A patient taking high-dose salicylate starts sodium bicarbonate regularly for heartburn. What happens to salicylate excretion, and why?

- A. Urinary excretion decreases, because the urine becomes alkaline B. Excretion is unchanged; antacids affect only gastric absorption
C. Excretion decreases, because Al^{3+} chelates salicylate in the gut D. Urinary excretion increases, because the urine becomes alkaline
E. Excretion increases, because the antacid induces CYP450

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — C — ECL cells (fundus) make, store and release histamine in response to ACh and gastrin; histamine then acts on parietal H_2 receptors.
2 — A — The lecture lists acute stress ulcer as an H_2 -antagonist use, usually given IV.
3 — E — Antacids raise pH; pepsin is inactive above pH 4, so its activity falls as acid falls.
4 — B — PGE_2/PGI_2 inhibit acid (via ↓ adenylate cyclase) and stimulate mucus and HCO_3^- and dilate mucosal vessels.
5 — D — Urine alkalinisation increases excretion of phenobarbital, methotrexate and salicylate.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pharmacology GIT Block · MEDucation by Mattin Majid · Source: Dr. Begard O. Berzinji lecture slides, SGL 1 (full deck)
Study alongside the lecture, not instead of it.